

GITHUB ASSIGNMENT

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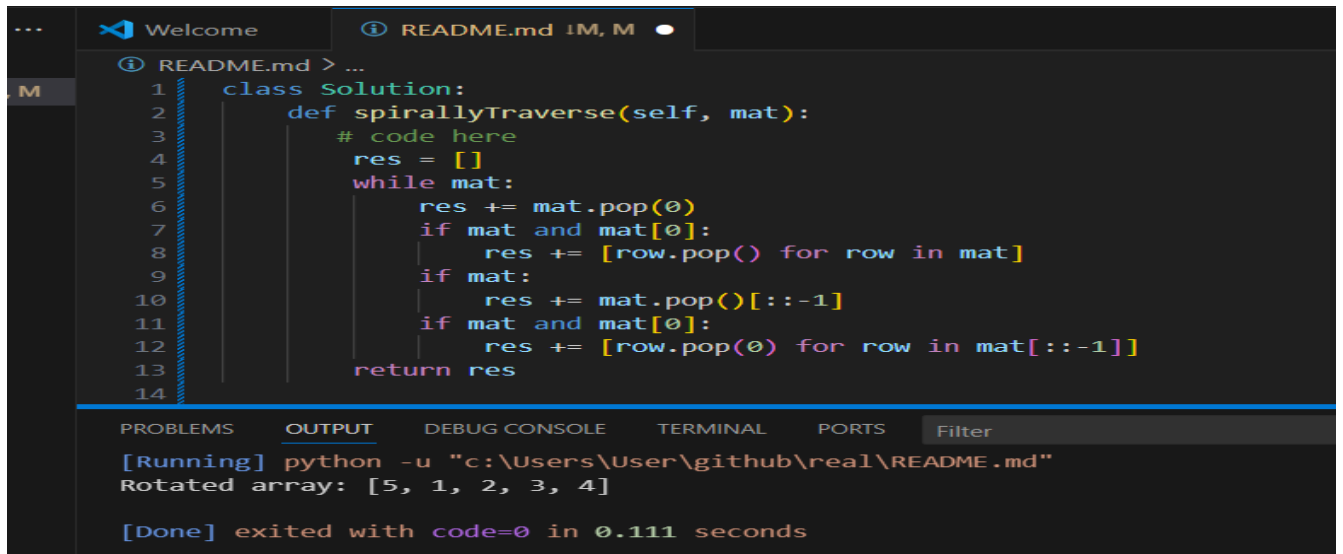
ASSIGNMENT ON PYTHON ND C BOTH

WEEK-2

Assignments (10 Problems):

1. Print matrix in spiral form.

PYTHON



```
... Welcome  README.md IM, M
M  README.md > ...
1 class Solution:
2     def spirallyTraverse(self, mat):
3         # code here
4         res = []
5         while mat:
6             res += mat.pop(0)
7             if mat and mat[0]:
8                 res += [row.pop() for row in mat]
9             if mat:
10                res += mat.pop()[::-1]
11            if mat and mat[0]:
12                res += [row.pop(0) for row in mat[::-1]]
13        return res
14
```

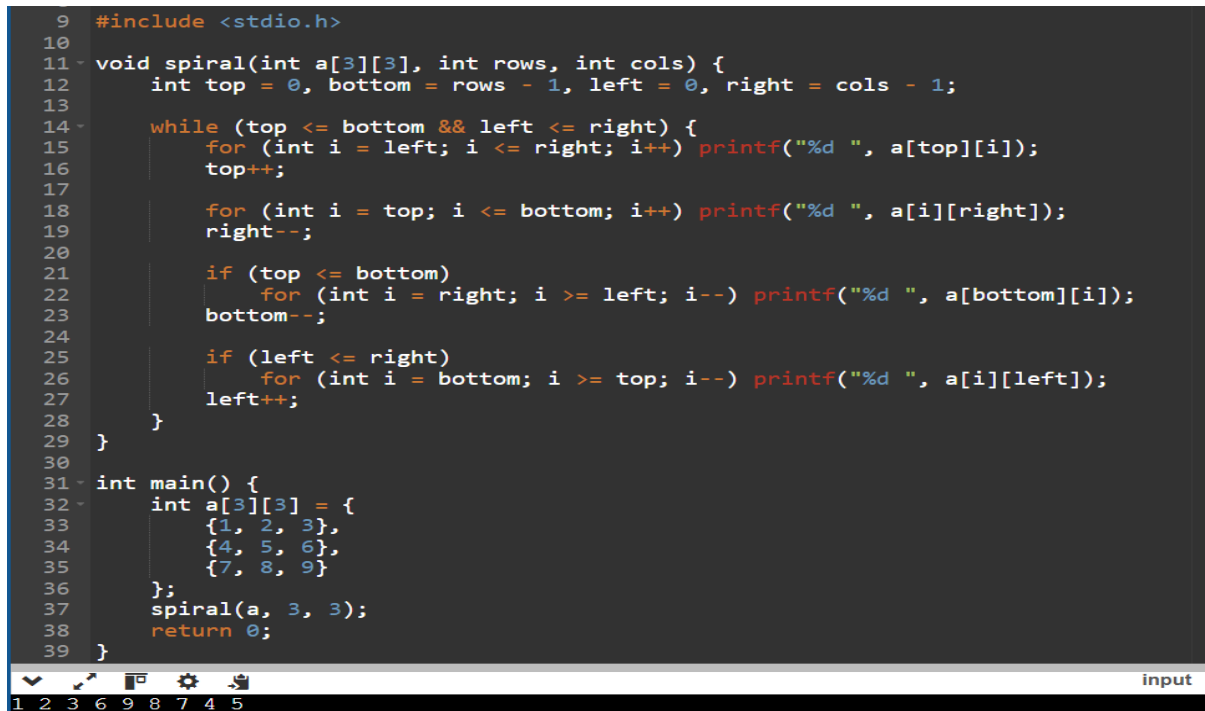
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS Filter

[Running] python -u "c:\Users\User\github\real\README.md"

Rotated array: [5, 1, 2, 3, 4]

[Done] exited with code=0 in 0.111 seconds

C



```
9 #include <stdio.h>
10
11 void spiral(int a[3][3], int rows, int cols) {
12     int top = 0, bottom = rows - 1, left = 0, right = cols - 1;
13
14     while (top <= bottom && left <= right) {
15         for (int i = left; i <= right; i++) printf("%d ", a[top][i]);
16         top++;
17
18         for (int i = top; i <= bottom; i++) printf("%d ", a[i][right]);
19         right--;
20
21         if (top <= bottom)
22             for (int i = right; i >= left; i--) printf("%d ", a[bottom][i]);
23         bottom--;
24
25         if (left <= right)
26             for (int i = bottom; i >= top; i--) printf("%d ", a[i][left]);
27         left++;
28     }
29 }
30
31 int main() {
32     int a[3][3] = {
33         {1, 2, 3},
34         {4, 5, 6},
35         {7, 8, 9}
36     };
37     spiral(a, 3, 3);
38     return 0;
39 }
```

input

1 2 3 6 9 8 7 4 5

2.Rotate matrix by 90 degree

Python

```
① README.md > ...
M
1 class Solution:
2     def rotateMatrix(self, mat):
3         n = len(mat)
4         for i in range(n):
5             for j in range(i, n):
6                 mat[i][j], mat[j][i] = mat[j][i], mat[i][j]
7         for j in range(n):
8             for i in range(n // 2):
9                 mat[i][j], mat[n - 1 - i][j] = mat[n - 1 - i][j], mat[i][j]
10    mat = [
11        [6, 3, 0],
12        [7, 4, 1],
13        [8, 5, 2]
14    ]
15    Solution().rotateMatrix(mat)
16    for row in mat:
17        print(row)

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

er/github/real/README.md
[0, 1, 2]
[3, 4, 5]
[6, 7, 8]
```

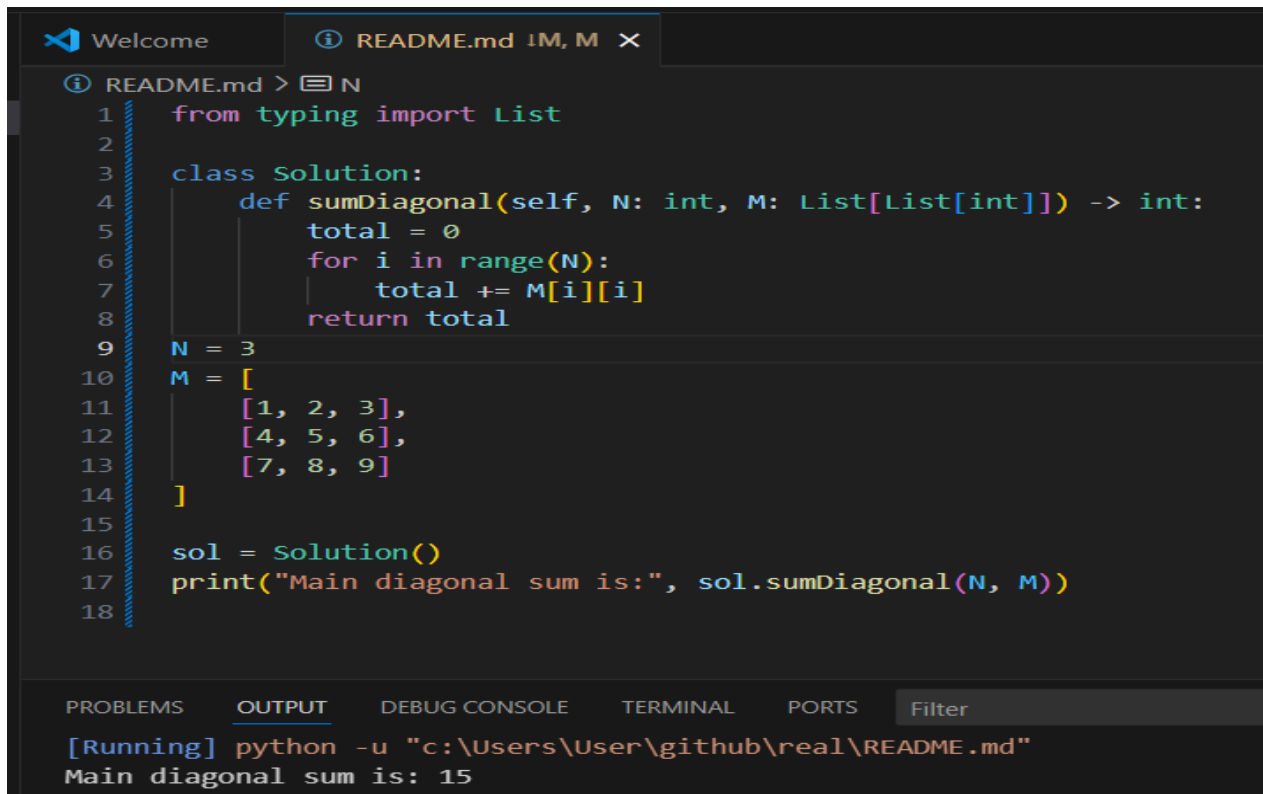
C

```
8
9
10 #include <stdio.h>
11
12 int main() {
13     int a[3][3] = {
14         {6, 3, 0},
15         {7, 4, 1},
16         {8, 5, 2}
17     };
18
19     // Directly print rotated result
20     for (int j = 2; j >= 0; j--) {
21         for (int i = 0; i < 3; i++) {
22             printf("%d ", a[i][j]);
23         }
24         printf("\n");
25     }
26
27     return 0;
28 }
```

```
0 1 2
3 4 5
6 7 8
```

3. Sum of diagonals.

Python



The screenshot shows a Python IDE with a file named 'README.md'. The code defines a class 'Solution' with a method 'sumDiagonal' that takes 'N' and 'M' as arguments. It calculates the sum of the main diagonal elements of the matrix 'M' by iterating from 0 to N-1 and adding 'M[i][i]' to a 'total' variable. Below the class definition, 'N' is set to 3 and 'M' is a 3x3 matrix with values [[1, 2, 3], [4, 5, 6], [7, 8, 9]]. A 'Solution' object 'sol' is created, and the result of 'sol.sumDiagonal(N, M)' is printed. The output at the bottom of the IDE shows 'Main diagonal sum is: 15'.

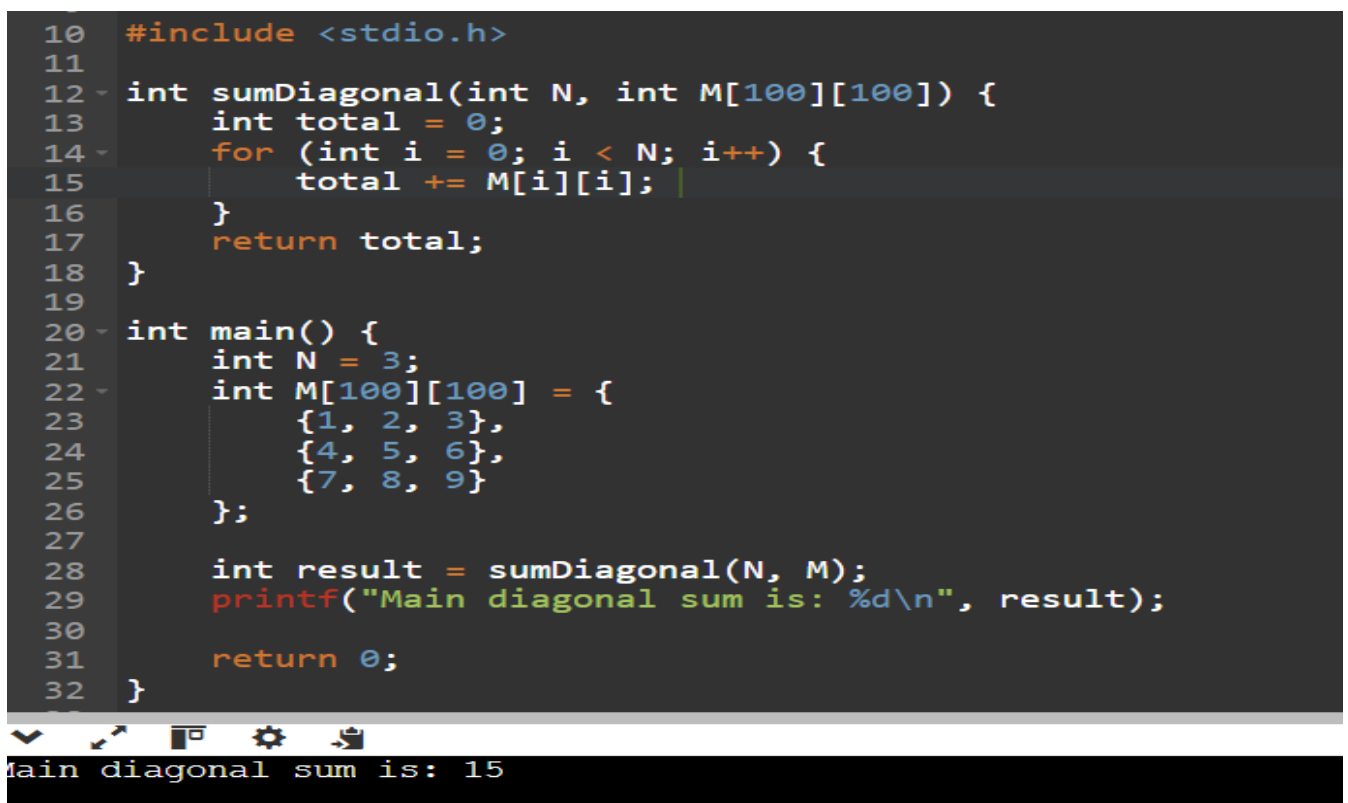
```
1 from typing import List
2
3 class Solution:
4     def sumDiagonal(self, N: int, M: List[List[int]]) -> int:
5         total = 0
6         for i in range(N):
7             total += M[i][i]
8         return total
9
10 N = 3
11 M = [
12     [1, 2, 3],
13     [4, 5, 6],
14     [7, 8, 9]
15 ]
16
17 sol = Solution()
18 print("Main diagonal sum is:", sol.sumDiagonal(N, M))
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS Filter

[Running] python -u "c:\Users\User\github\real\README.md"

Main diagonal sum is: 15

C



The screenshot shows a C IDE with a code editor and a terminal. The code includes 'stdio.h' and defines a function 'sumDiagonal' that takes 'N' and a 2D array 'M' as arguments. It calculates the sum of the main diagonal elements by iterating from 0 to N-1 and adding 'M[i][i]' to a 'total' variable. In the 'main' function, 'N' is set to 3 and 'M' is a 3x3 matrix with values {{1, 2, 3}, {4, 5, 6}, {7, 8, 9}}. The 'sumDiagonal' function is called, and the result is printed using 'printf'. The terminal output shows 'Main diagonal sum is: 15'.

```
10 #include <stdio.h>
11
12 int sumDiagonal(int N, int M[100][100]) {
13     int total = 0;
14     for (int i = 0; i < N; i++) {
15         total += M[i][i];
16     }
17     return total;
18 }
19
20 int main() {
21     int N = 3;
22     int M[100][100] = {
23         {1, 2, 3},
24         {4, 5, 6},
25         {7, 8, 9}
26     };
27
28     int result = sumDiagonal(N, M);
29     printf("Main diagonal sum is: %d\n", result);
30
31     return 0;
32 }
```

Main diagonal sum is: 15

4. Transpose of matrix

Python

```
1 class Solution:
2     def transpose(self, mat):
3         return [[mat[i][j] for i in range(len(mat))] for j in range(len(mat[0]))]
4 sol = Solution()
5 matrix = [
6     [1, 2, 3],
7     [4, 5, 6]
8 ]
9 result = sol.transpose(matrix)
10
11 print("Transpose is:")
12 for row in result:
13     print(*row)
14
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
3 6
PS C:\Users\User\github\real> & C:/Users/User/AppData/Local/Programs/Python/Python313/python3.13t.exe c:/Users/Us
er/github/real/README.md
Transpose is:
1 4
2 5
3 6
PS C:\Users\User\github\real>
```

C

```
9
10 #include <stdio.h>
11
12 int main() {
13     int mat[2][3] = {
14         {1, 2, 3},
15         {4, 5, 6}
16     };
17
18     int rows = 2, cols = 3;
19
20     printf("Transpose is:\n");
21     for (int j = 0; j < cols; j++) {
22         for (int i = 0; i < rows; i++) {
23             printf("%d ", mat[i][j]);
24         }
25         printf("\n");
26     }
27
28     return 0;
29 }
30
31
```

Transpose is:

```
1 4
2 5
3 6
```

5. Check if matrix is sparse

Python

```
① README.md > is_sparse
1 def is_sparse(matrix):
2     rows = len(matrix)
3     cols = len(matrix[0])
4     total = rows * cols
5
6     zero_count = sum(row.count(0) for row in matrix)
7
8     return zero_count > (total // 2)
9
10 mat = [
11     [0, 0, 3],
12     [0, 5, 0],
13     [0, 0, 0]
14 ]
15
16 if is_sparse(mat):
17     print("Matrix is sparse")
18 else:
19     print("Matrix is not sparse")
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

er/github/real/README.md
Matrix is sparse
PS C:\Users\User\github\real>

C

```
9
10 #include <stdio.h>
11
12 int main() {
13     int mat[3][3] = {
14         {0, 0, 3},
15         {0, 5, 0},
16         {0, 0, 0}
17     };
18
19     int rows = 3, cols = 3;
20     int zeroCount = 0, total = rows * cols;
21     for (int i = 0; i < rows; i++) {
22         for (int j = 0; j < cols; j++) {
23             if (mat[i][j] == 0)
24                 zeroCount++;
25         }
26     }
27     if (zeroCount > total / 2)
28         printf("Matrix is sparse\n");
29     else
30         printf("Matrix is not sparse\n");
31     return 0;
32 }
33
34
```

input
Matrix is sparse

6. Add two polynomials

Python

```
① README.md > ...
1  poly1 = [5, 2, 3]
2  poly2 = [6, 4, -1]
3  degree = 2
4
5  sum_poly = [poly1[i] + poly2[i] for i in range(degree + 1)]
6
7  print("Sum:", end=" ")
8  for i in range(degree, -1, -1):
9      if sum_poly[i] != 0:
10         if i != degree:
11             print("+", end=" ")
12             print(f"{sum_poly[i]}x^{i}", end=" ")
13
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

Sum: 2x^2 + 6x^1 + 11x^0
PS C:\Users\User\github\real> |

C

```
9
10 #include <stdio.h>
11
12 int main() {
13     int poly1[10] = {5, 2, 3};
14     int poly2[10] = {6, 4, -1};
15     int sum[10];
16     int degree = 2;
17
18     for (int i = 0; i <= degree; i++)
19         sum[i] = poly1[i] + poly2[i];
20
21     printf("Sum: ");
22     for (int i = degree; i >= 0; i--) {
23         if (sum[i] != 0) {
24             if (i != degree) printf(" + ");
25             printf("%dx^%d", sum[i], i);
26         }
27     }
28
29     return 0;
30 }
31
```

Sum: 2x^2 + 6x^1 + 11x^0

...Program finished with exit code 0
Press ENTER to exit console.

7. Multiply two polynomials

Python

```
... Welcome README.md IM, M X
① README.md > ...
IM, M
1 a = [1, 2, 3]
2 b = [4, 5]
3 m, n = len(a), len(b)
4 result = [0] * (m + n - 1)
5
6 for i in range(m):
7     for j in range(n):
8         result[i + j] += a[i] * b[j]
9
10 print("Result:", end=" ")
11 for i in range(len(result)):
12     if result[i] != 0:
13         if i == 0:
14             print(f"{result[i]}", end="")
15         else:
16             print(f" + {result[i]}x^{i}", end="")
17
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
PS C:\Users\User\github\real>
> & C:/Users/User/AppData/Local/Programs/Python/Python
er/github/real/README.md
Result: 4 + 13x^1 + 22x^2 + 15x^3
PS C:\Users\User\github\real> & C:/Users/User/AppData/Local/Programs/Python/Python
```

C

```
9 #include <stdio.h>
1
2 int main() {
3     int a[] = {1, 2, 3};
4     int b[] = {4, 5};
5     int m = 3, n = 2;
6     int result[m + n - 1];
7     for (int i = 0; i < m + n - 1; i++)
8         result[i] = 0;
9     for (int i = 0; i < m; i++)
10         for (int j = 0; j < n; j++)
11             result[i + j] += a[i] * b[j];
12     printf("Result: ");
13     for (int i = 0; i < m + n - 1; i++) {
14         if (result[i] != 0) {
15             if (i == 0)
16                 printf("%d", result[i]);
17             else
18                 printf(" + %dx^%d", result[i], i);
19         }
20     }
21     return 0;
22 }
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8. Identity matrix check

Python

```
① README.md > ...
1  matrix = [
2      [1, 0, 0],
3      [0, 1, 0],
4      [0, 0, 1]
5  ]
6
7  n = len(matrix)
8  is_identity = True
9
10 for i in range(n):
11     for j in range(n):
12         if (i == j and matrix[i][j] != 1) or (i != j and matrix[i][j] != 0):
13             is_identity = False
14             break
15     if not is_identity:
16         break
17
18 print("Identity matrix" if is_identity else "Not an identity matrix")
19
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

Identity matrix
PS C:\Users\User\github\real>

C

```
10 #include <stdio.h>
11
12 int main() {
13     int matrix[3][3] = {
14         {1,0,0},
15         {0,1,0},
16         {0,0,1}
17     };
18
19     for (int i = 0; i < 3; i++)
20         for (int j = 0; j < 3; j++)
21             if ((i == j && matrix[i][j] != 1) || (i != j && matrix[i][j] != 0)) {
22                 printf("Not an identity matrix\n");
23                 return 0;
24             }
25
26     printf("Identity matrix\n");
27     return 0;
28 }
29
```

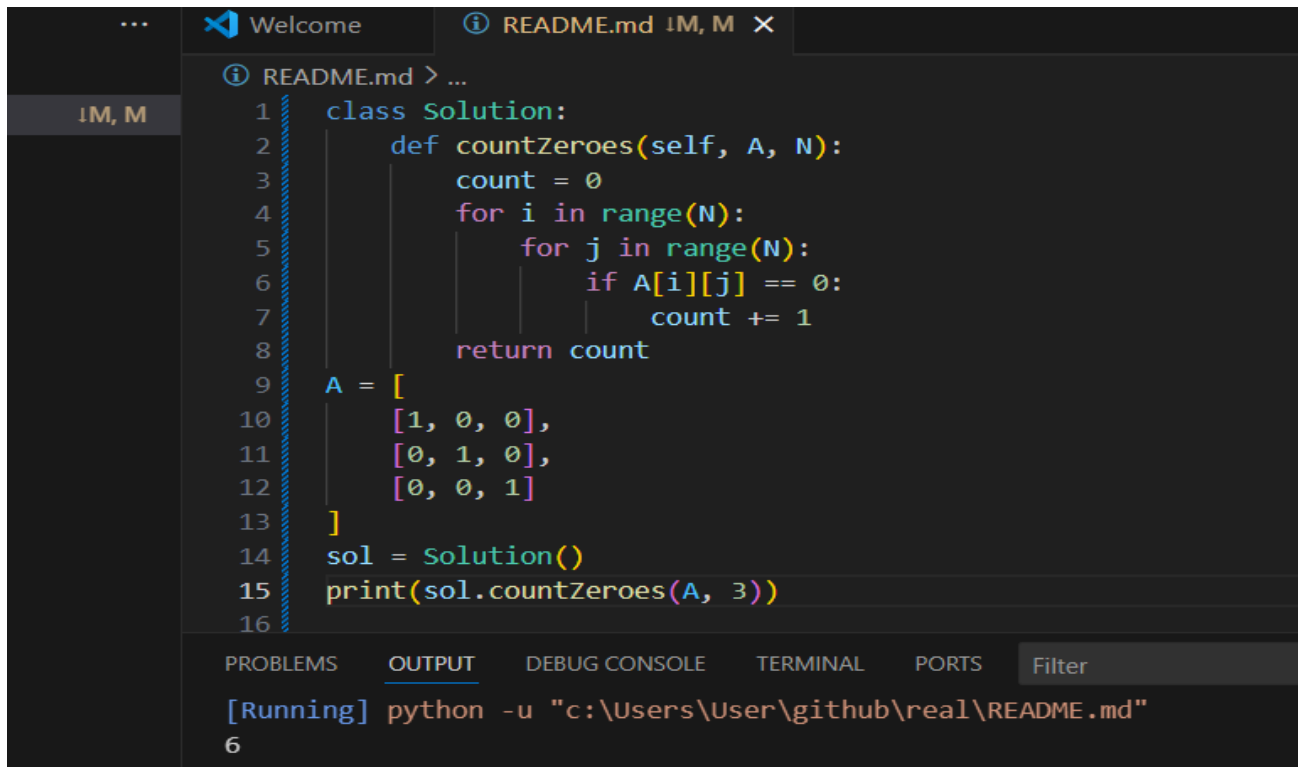
input

Identity matrix

...Program finished with exit code 0
Press ENTER to exit console.

9. Count zeros in matrix

Python



```
class Solution:
    def countZeroes(self, A, N):
        count = 0
        for i in range(N):
            for j in range(N):
                if A[i][j] == 0:
                    count += 1
        return count

A = [
    [1, 0, 0],
    [0, 1, 0],
    [0, 0, 1]
]

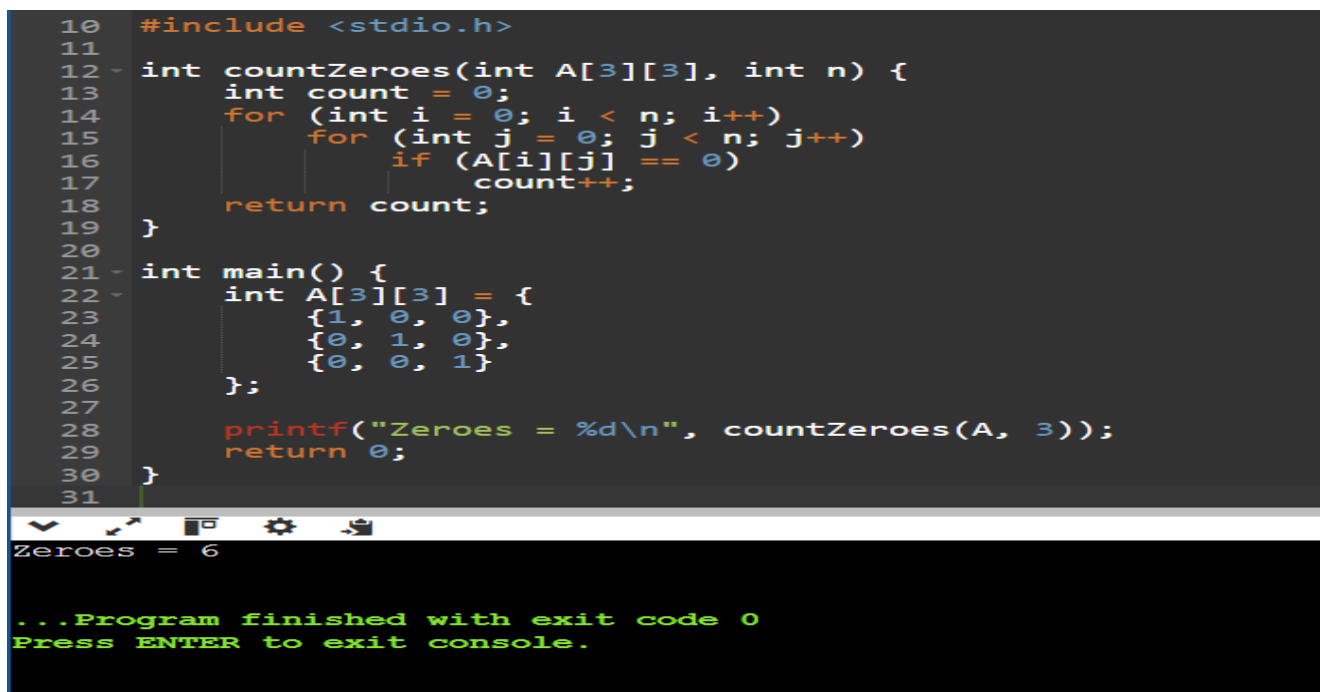
sol = Solution()
print(sol.countZeroes(A, 3))
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS Filter

[Running] python -u "c:\Users\User\github\real\README.md"

6

C



```
#include <stdio.h>

int countZeroes(int A[3][3], int n) {
    int count = 0;
    for (int i = 0; i < n; i++)
        for (int j = 0; j < n; j++)
            if (A[i][j] == 0)
                count++;
    return count;
}

int main() {
    int A[3][3] = {
        {1, 0, 0},
        {0, 1, 0},
        {0, 0, 1}
    };

    printf("Zeroes = %d\n", countZeroes(A, 3));
    return 0;
}
```

Zeroes = 6

...Program finished with exit code 0
Press ENTER to exit console.

=====END=====